Chapter 6: Degree Theory, k-Set Contractions and Condensing Operators

Nonlinear Analysis and Fixed Point Theory

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6.2.1 Invariance Principles in Degree Theory

Challenge: When defining a completely continuous vector field $\Phi = I - A$ in infinite-dimensional Banach spaces, the choice of Banach space is crucial. The geometric properties depend heavily on this choice.

Question: If $\Phi = I - A$ is completely continuous in two different Banach spaces X_1 and X_2, how do the rotation numbers $\gamma(\Phi, \Omega_i, 0)$ relate?

Key Definitions:

- Common core: Sets $\Omega_1 \subset X_1$ and $\Omega_2 \subset X_2$ have a common core if the fixed points of A in the closures of Ω_1 and Ω_2 coincide
- Compatible norms: When embedding spaces, the norm structures preserve key properties
- Condition $\alpha_0(\partial\Omega)$: Technical condition ensuring continuity properties of the vector field on the boundary

Main Invariance Principles

Theorem (6.19 - Invariance Theorem)

Let X_1 and X_2 be Banach spaces with $\Omega_1 \subset X_1$, $\Omega_2 \subset X_2$ bounded open sets. Assume $\Phi = I - A$ is completely continuous on Ω_1 and Ω_2 with:
Case (a): *A continuous on $\overline{\Omega}_1, \overline{\Omega}_2$ as an operator $X_i \to X_i$.*
Case (b): *Norms are compatible; A is locally bounded and compact.*
Case (c): *Norms compatible; $\exists$ compact convex sets $S_1 \subset X_1 \cap X_2$, with $A(S_1 \cap \Omega_1) \subset S_2$ and $A(S_2 \cap \Omega_2) \subset S_1$.*
Then: $\gamma(\Phi, \Omega_1) = \gamma(\Phi, \Omega_2)$

Interpretation: The degree is independent of the choice of Banach space (under suitable conditions).

Computing the Index: If Ω is bounded and $N(\Phi, \Omega)$ (zeros of Φ in $\overline{\Omega}$) is finite $\{x_0, \ldots, x_s\}$:

$$\gamma(\Phi, \Omega) = \operatorname{ind}(x_0, \Phi) + \cdots + \operatorname{ind}(x_s, \Phi)$$

where $\operatorname{ind}(x_i, \Phi)$ is the local index at an isolated zero.

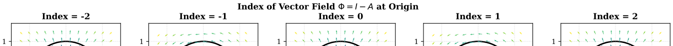
Example (Equation 6.13):

```
# Python: Computing index for vector field at origin
import numpy as np

def vector_field(x, y):
    """Phi(x,y) = (x - f1(x,y), y - f2(x,y))"""
    return np.array([x - 0.1*x*y, y + 0.05*x**2])

# Evaluate on boundary of Omega
theta = np.linspace(0, 2*np.pi, 1000)
x_boundary = 2 * np.cos(theta)
y_boundary = 2 * np.sin(theta)
phi_vals = [vector_field(x, y) for x, y in zip(x_boundary, y_boundary)]

# Index ~ winding number around origin
```



6.3 Introduction to Skrypnik Degree Theory

Skrypnik's 1973 monograph developed a new topological degree theory for mappings:

- From a reflexive space X to its dual X^*
- Class α operators (including demicontinuous and pseudomonotone)
- Applications to solvability of nonlinear elliptic PDEs

Important cases:

1. $\Gamma = X^*$ (general case)
2. X reflexive, $\Gamma = X^*$ (standard)
3. $X = Z^*$ where Z is Banach, $\Gamma = Z \subseteq X^{**}$

Key technical aspect: For operators $\Phi : \mathfrak{M} \to X^*$ with $\mathfrak{M} \subset X$:

- $\langle u, v \rangle$ denotes the action of the functional $u \in X^*$ on $v \in X$
- Weak and strong convergence denoted by $\rightharpoonup$ and $\to$ respectively

Definition of Class α Operators

Definition (Class α - Definition 6.6)

An operator $\Phi : \mathfrak{M} \to X^*$ satisfies condition $\alpha(\mathfrak{M})$ if:

$$\limsup_{n\to\infty} \langle \Phi(x_n), x_n - x_0 \rangle \leq 0$$

for any sequence $x_n \in \mathfrak{M}$ with $x_n \rightharpoonup x_0$.
This is satisfied when:

- Φ is **pseudomonotone**: If $x_n \rightharpoonup x_0$ and $\limsup \langle \Phi(x_n), x_n - x_0 \rangle \leq 0$, then for all v:

$$\liminf_{n\to\infty} \langle \Phi(x_n), x_n - v \rangle \geq \langle \Phi(x_0), x_0 - v \rangle$$

- Φ is **demicontinuous** and monotone
- Φ is **completely continuous** on reflexive spaces

Skrypnik Degree Definition

Theorem (6.22 - Skrypnik Degree Definition)

Let Ω be a bounded open set in reflexive Banach space X, and $A : \bar{\Omega} \to X^*$ a demicontinuous pseudomonotone operator with:

$$\|Au\|_{X^*} \geq \delta_0 > 0 \quad \text{for all } u \in \partial\Omega$$

If $0 \notin A(\partial\Omega)$, the Skrypnik degree

$$\deg(A, \Omega, 0) = \lim_{n\to\infty} \deg(A_n, \Omega_n, 0)$$

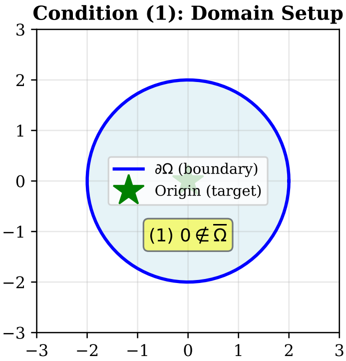
is well-defined, where $A_n = P_n A|_{F_n}$ are finite-dimensional restrictions.

Key properties:

- The degree exists and is independent of the finite-dimensional approximation
- Satisfies homotopy invariance under suitable conditions
- $\deg(A, \Omega, 0) \neq 0 \Rightarrow$ existence of solutions to $Au = u$ in Ω

Skrypnik Degree Conditions

Skrypnik's Degree Theory: $\Phi = I - A$, $A : \bar{\Omega} \to X^*$



Condition (2): Operator A properties

- Completely continuous on $\overline{\Omega}$
- Pseudomonotone or demicontinuous
- $\|Au\|_{X^*} \geq \delta_0 > 0$ for $u \in \partial\Omega$

These ensure $\Phi = I - A$ satisfies

the $\alpha(\partial\Omega)$ condition

Skrypnik's Theorem: If conditions (1) and (2) hold, then $\deg(A, \overline{\Omega}, 0)$ **is well-defined and possesses properties similar to finite-dimensional degree. Key properties:**

Key Lemma (for proving Skrypnik's theorem):
When approximating A by finite-dimensional restrictions A_n on subspaces F_n:

$$\lim_{n\to\infty} \deg(A_n, \Omega_n, 0) = \deg(A, \Omega, 0) \quad \text{(independent of sequence)} \tag{1}$$

Proof strategy:

1. Use the fact that finite-dimensional degree is well-established
2. $[tA_n + (1-t)\tilde{A}_n]u \neq 0$ for $u \in \partial\Omega_n$, $t \in [0,1]$ (homotopy condition)
3. As $n \to \infty$, the degress stabilize to the infinite-dimensional degree

Remark 6.7: The Skrypnik degree shares essential properties with the Leray-Schauder degree but applies to a broader class of operators (pseudomonotone, not just compact).

6.4.1 k-Set Contractions: Definition and Properties

Definition (k-Set Contraction)

A mapping $T : X \to X$ is a k-set contraction if:

$$\mu(T(A)) \leq k \cdot \mu(A)$$

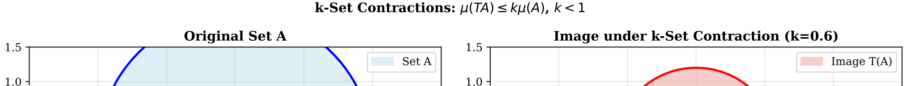
for every bounded set $A \subseteq X$, where μ is the **measure of noncompactness** and $k \in [0, 1)$.

Special cases:

- $k = 0$: **Compact operator** (image has measure zero)
- $k = 1$: **Nonexpansive mapping** in the measure sense
- $0 < k < 1$: Strict contraction on the measure of noncompactness

Example 6.4: The measure of noncompactness can be defined as:

$$\mu(A) = \inf\{\epsilon > 0 : A \text{ can be covered by finitely many balls of radius } \epsilon\}$$



Measure of Noncompactness: Kuratowski Measure

Definition (Kuratowski Measure of Noncompactness)

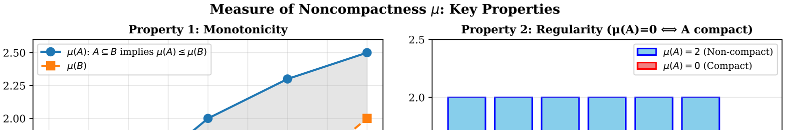
For a bounded set A in a metric space (X, d):

$$\alpha(A) = \inf \{\epsilon > 0 : A \text{ has a finite covering by sets of diameter } \leq \epsilon\}$$

Key properties:

1. **Monotonicity**: $A \subseteq B \Rightarrow \alpha(A) \leq \alpha(B)$
2. **Regularity**: $\alpha(A) = 0 \Leftrightarrow A$ is *compact*
3. **Additivity**: $\alpha(A \cup B) = \max\{\alpha(A), \alpha(B)\}$
4. **Subadditivity**: $\alpha(A + B) \leq \alpha(A) + \alpha(B)$ (Minkowski sum)
5. **Homogeneity**: $\alpha(\lambda A) = |\lambda|\alpha(A)$

Hausdorff measure: Related but defined via ϵ-coverings in terms of the number of balls.



Properties of Measure of Noncompactness

Fundamental Theorem: For any bounded set A and continuous operator T:

$$\alpha(T(A)) \leq \text{diam}(T(A))$$

Theorem 6.23 (Darbo): If $T : \Omega \to \Omega$ is continuous on bounded closed convex set Ω with:

$$\mu(T(X)) \leq k \cdot \mu(X) \quad \text{for some } k < 1$$

then T has a fixed point in Ω.

Code Example: Computing Measure of Noncompactness

```python
# Estimate measure of noncompactness via covering
def measure_noncompactness(points, epsilon_tolerance=0.01):
    """Compute approximate Kuratowski measure"""
    from scipy.spatial.distance import pdist, squareform

    dists = squareform(pdist(points))
    # Greedy covering algorithm
    covered = set()
    num_balls = 0
    for i in range(len(points)):
        if i not in covered:
            num_balls += 1
            # Find all points within epsilon
```

Darbo's Fixed Point Theorem

Theorem (6.24 - Darbo)

Let Ω be a nonempty, bounded, closed and convex subset of a Banach space E, and let $T : \Omega \to \Omega$ be a continuous mapping. If there exists a constant $k \in [0, 1)$ such that:

$$\mu(T(X)) \leq k \cdot \mu(X)$$

for any nonempty subset $X \subseteq \Omega$, where μ is the measure of noncompactness, then T has a fixed point in Ω.

Key observations:

1. Generalizes Banach Contraction Principle to infinite dimensions
2. Does not require T to be a contraction in the usual sense ($\|Tx - Ty\| \leq k\|x - y\|$)
3. Uses measure of noncompactness instead of pointwise distance

Remark: If K is compact or F is compact, Theorem 5.24 (Schauder) is a special case. If K is compact and F is weakly continuous in reflexive Banach space, this is the Tychonoff fixed point theorem.

Nonlinear k-Set Contractions

Definition (Nonlinear k-Set Contraction)

A mapping $T : X \to X$ is a nonlinear k-set contraction if there exists a continuous $\varphi : \mathbb{R}^+ \to \mathbb{R}^+$ with $\varphi(0) = 0$ such that:

$$\mu(T(A)) \leq \varphi(\mu(A)) \quad \text{for all bounded } A \subseteq X$$

where $\varphi(r) < r$ for all $r > 0$.

Examples of φ functions:

- **Linear**: $\varphi(r) = kr$ with $k < 1$ (standard k-set contraction)
- $\varphi(r) = \frac{kr}{1+r}$ with $L > 0$, $K > 0$
- $\varphi(r) = \log(1 + r)$

Lemma (6.9 - Limit Behavior)

If φ is a φ-function on $\mathbb{R}^+$ into itself with $\varphi(r) < r$ for $r > 0$, then:

$$\lim_{n\to\infty} \varphi^n(t) = 0 \quad \textit{for all } t \in [0, \infty)$$

Condensing Operators: Definition

Definition (α-Condensing Operator)

A continuous mapping $T : \Omega \to X$ is called α-condensing if for every nonempty bounded set $A \subseteq \Omega$:

$$\alpha(T(A)) < \alpha(A)$$

where α is the measure of noncompactness, **unless** A is compact (in which case $\alpha(A) = 0$).

Intuition: A condensing operator strictly reduces the measure of noncompactness of any noncompact set. This is a refined notion between k-set contractions and completely continuous operators.

Relationship to other classes:

- Every compact operator is condensing
- Every k-set contraction with $k < 1$ is condensing
- Not all condensing operators are k-set contractions
- Condensing operators can be weakly continuous

Theorem (Sadovskii - Generalized Fixed Point Theorem)

Let Ω be a nonempty, bounded, closed and convex subset of a Banach space X, and let $T : \Omega \to \Omega$ be a continuous mapping. If T is α-condensing (where α is the measure of noncompactness), then T has at least one fixed point in Ω.

Comparison with previous theorems:

Theorem	Requirement	Operator Type	Conclusion
Schauder	Compact K	Continuous	FP exists
Banach	$\|Tx - Ty\| \leq k\|x - y\|$	Contraction	Unique FP
Darbo	$\mu(T(X)) \leq k\mu(X)$	k-set contract.	FP exists
Sadovskii	$\alpha(T(X)) < \alpha(X)$	Condensing	FP exists

Definition 6.18: Dhage's Concept

Definition (D-Set-Lipschitz Mapping)

A mapping $T : X \to X$ is called D-set-Lipschitz if there exists an upper semicontinuous nondecrasing function $\varphi : \mathbb{R}^+ \to \mathbb{R}^+$ with $\varphi(0) = 0$ such that:

$$\mu(T(A)) \leq \varphi(\mu(A))$$

for all $A \in \mathcal{B}(X)$ with $T(A) \subseteq B(X)$, where $\varphi(r) = kr$ defines a **k-set Lipschitz mapping** ($k > 0$). If $k < 1$, then T is a **k-set contraction**; if $\varphi(r) < r$ for $r > 0$, it is a **nonlinear D-set contraction**.

Remark 6.7 (Dhage): Properties of D-functions:

- If φ, ψ are D-functions, then *i)* $\varphi + \psi$, *ii)* $\lambda\varphi$ ($\lambda > 0$), and *iii)* $\varphi \circ \psi$ are also D-functions
- Common D-functions: $\psi(r) = kr$, $\psi(r) = \frac{kr}{1+r}$, $\psi(r) = \log(1 + r)$

Theorem 6.28-6.29: Darbo-type Results

Theorem (6.28 - Darbo's Generalization)

Let Ω be a nonempty, bounded, closed and convex subset of a Banach space E, and let $T : \Omega \to \Omega$ be a continuous mapping. If there exists a constant $k \in [0, 1)$ such that:

$$\mu(T(X)) \leq k\mu(X)$$

for any nonempty subset X of Ω (where μ is the measure of noncompactness), then T has a fixed point in Ω.

Theorem (6.29 - Nonlinear D-Set Contraction (Dhage))

Let C be a closed, convex and bounded subset of a Banach space X, and let $T : C \to C$ be a continuous and nonlinear D-set contraction. Then T has a fixed point.

Proof idea: Use the property that for a nonlinear D-function with $\varphi(r) < r$ for $r > 0$, the iterates $\varphi^n(t) \to 0$ as $n \to \infty$.

Remark 6.8 - Attractivity of Fixed Points:
If $\tilde{F}(T) = \{x \in C : Tx = x\}$ denotes the set of fixed points of $T : C \to C$, then:

1. $\tilde{F}(T) \in \ker \mu$ (the set of fixed points has measure zero, i.e., is compact)
2. If $\tilde{F}(T) \notin \ker \mu$, then $\mu(\tilde{F}(T)) > 0$ and we have $\mu(T(\tilde{F}(T))) = \mu(\tilde{F}(T))$
3. From nonlinear set contractivity: $\mu(T(\tilde{F}(T))) \leq \varphi(\mu(\tilde{F}(T)))$
4. Since $\varphi(r) < r$ for $r > 0$, this is a **contradiction**
5. Therefore: $\tilde{F}(T) \in \ker \mu$ (i.e., $\tilde{F}(T)$ is compact)

This particular property of measures has important applications to the **attractivity of solutions** in the study of nonlinear functional integral equations.

Example: Fixed Point via k-Set Contraction

Problem: Find fixed points of $T : L^2[0,1] \to L^2[0,1]$ defined by:

$$(Tx)(t) = \int_0^1 K(t,s) f(s, x(s))\, ds$$

where K is a kernel and f is a Carathéodory function.

Approach using Darbo's theorem:

```python
# Verify k-set contraction property
def verify_k_set_contraction(T, measured_sets, k=0.7):
    """
    Check if T is k-set contraction by estimating
    measure of noncompactness of image
    """
    results = []
    for A in measured_sets:
        TA = [T(x) for x in A]
        mu_A = measure_noncompactness(A)
        mu_TA = measure_noncompactness(TA)

        ratio = mu_TA / mu_A if mu_A > 0 else 0
        results.append({
            'set': A, 'mu_A': mu_A, 'mu_TA': mu_TA,
            'ratio': ratio, 'is_k_contract': ratio <= k
        })
```

Applications to Integral and Differential Equations

Application 1: Nonlinear Integral Equations

- Fredholm equations: $x(t) = f(t) + \int_a^b K(t,s)g(s,x(s))\,ds$
- Using k-set contractions and condensing operators to establish existence

Application 2: Evolution Equations in Banach Spaces

- Partial differential equations with nonlinear lower-order terms
- Skrypnik degree theory for pseudomonotone operators $A : H^1 \to H^{-1}$

Application 3: Fixed Point Sequences

- When fixed points are not unique, study the structure of $\tilde{F}(T) = \{x : Tx = x\}$
- Attractivity properties: if $\tilde{F}(T)$ is nonempty, it must be compact

Application 4: Variational Inequalities

- Find $x \in K$ such that $\langle Ax, v - x \rangle \geq 0$ for all $v \in K$
- Use Skrypnik degree or condensing operator theory

Summary and Key Takeaways

Chapter 6 Main Results:

1. **Invariance Principles (6.2):** Degree theory is independent of Banach space choice under suitable conditions
2. **Skrypnik Degree (6.3):** Extends finite-dimensional degree to pseudomonotone operators on reflexive spaces
   - Handles operators mapping $X \to X^*$
   - Based on finite-dimensional approximations
3. **k-Set Contractions (6.4):** Use measure of noncompactness μ instead of pointwise distances
   - Darbo's Theorem: $\mu(T(X)) \leq k\mu(X)$ with $k < 1 \Rightarrow$ fixed point exists
   - Generalizes Banach Contraction Principle
4. **Condensing Operators (6.5):** Even broader class than k-set contractions
   - Sadovskii's theorem: condensing operators have fixed points
   - Applications to differential and integral equations

Connection to Previous Chapters

Fixed Point Theorems Hierarchy:

1. **Chapter 2:** Banach Contraction Mapping (complete metric spaces, unique FP)
2. **Chapter 3:** Nonexpansive and quasi-nonexpansive mappings
3. **Chapter 4:** Iterative methods (Mann, Ishikawa, Noor iterations)
4. **Chapter 5:** Schauder, Brouwer fixed point theorems (topological methods)
5. **Chapter 6:** Degree theory, condensing operators, measure-theoretic approaches

Progression: Pointwise distance $\rightarrow$ Topological degree $\rightarrow$ Measure of noncompactness $\rightarrow$ Condensing operators

Future chapters: These tools apply to variational inequalities, differential equations, and integral equations (Chapters 8–9).

References and Further Reading

Key papers cited in Chapter 6:

- **[564]** Skrypnik, I.V., 1973. Nonlinear elliptic equations of higher order
- **[155]** Darbo, G., 1955. Punti uniti in trasformazioni a codominio non compatto
- **[175]** Dhage, B.C., Various works on D-set-Lipschitz mappings and fixed points
- **[176]** Dhage, B.C., Properties of D-functions in condensing operators

Recommended further study:

1. Degree theory in infinite dimensions: See Chapter 6.2–6.3
2. k-Set contractions: Darbo (1955), generalized by many
3. Measure of noncompactness: Kuratowski, Hausdorff measures
4. Applications: See Chapters 8 (differential equations), 9 (integral equations)